TestBoard_7953
Final 20 / 30 MHz comparison

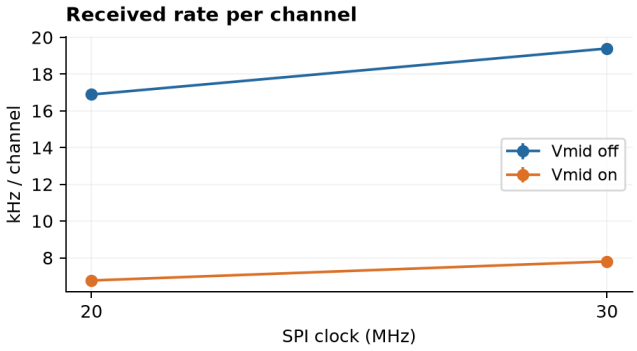
13 October 2026 | Production firmware | Manual conversion mode | LPSPI engine

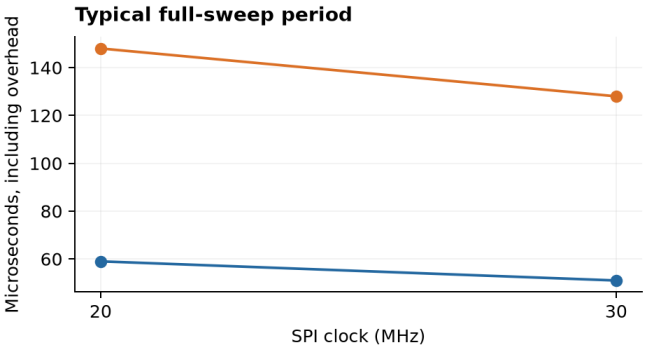
Setup and sampling rate

Both full arrays scanned in ADC order: 50 channels per sweep. Array 1 connector disconnected; Array 2 sensor strip attached and untouched. Laptop charger connected. Channel repeat = 1; Vref = 2.5 V; optional Vmid insertion on/off. Mandatory Vmid parking remains active in all captures.

Three repetitions per configuration; each capture has 1 s warm-up plus 15 s measurement. Configuration order rotates between repetitions. The first second is excluded from reported rate and noise. The board was not reflashed; its ordinary production status was checked, with profiling unavailable/disabled.

SPI / Vmid	Received rate Hz / channel	Total values/s (50 channels)	Sweep time us	Period incl. overhead, us	Overhead us
20 MHz / off	16,887.1	844,354	58.0	59.0	1.0
20 MHz / on	6,763.7	338,187	146.0	148.0	2.0
30 MHz / off	19,391.8	969,588	50.0	51.0	1.0
30 MHz / on	7,799.9	389,994	127.0	128.0	1.0





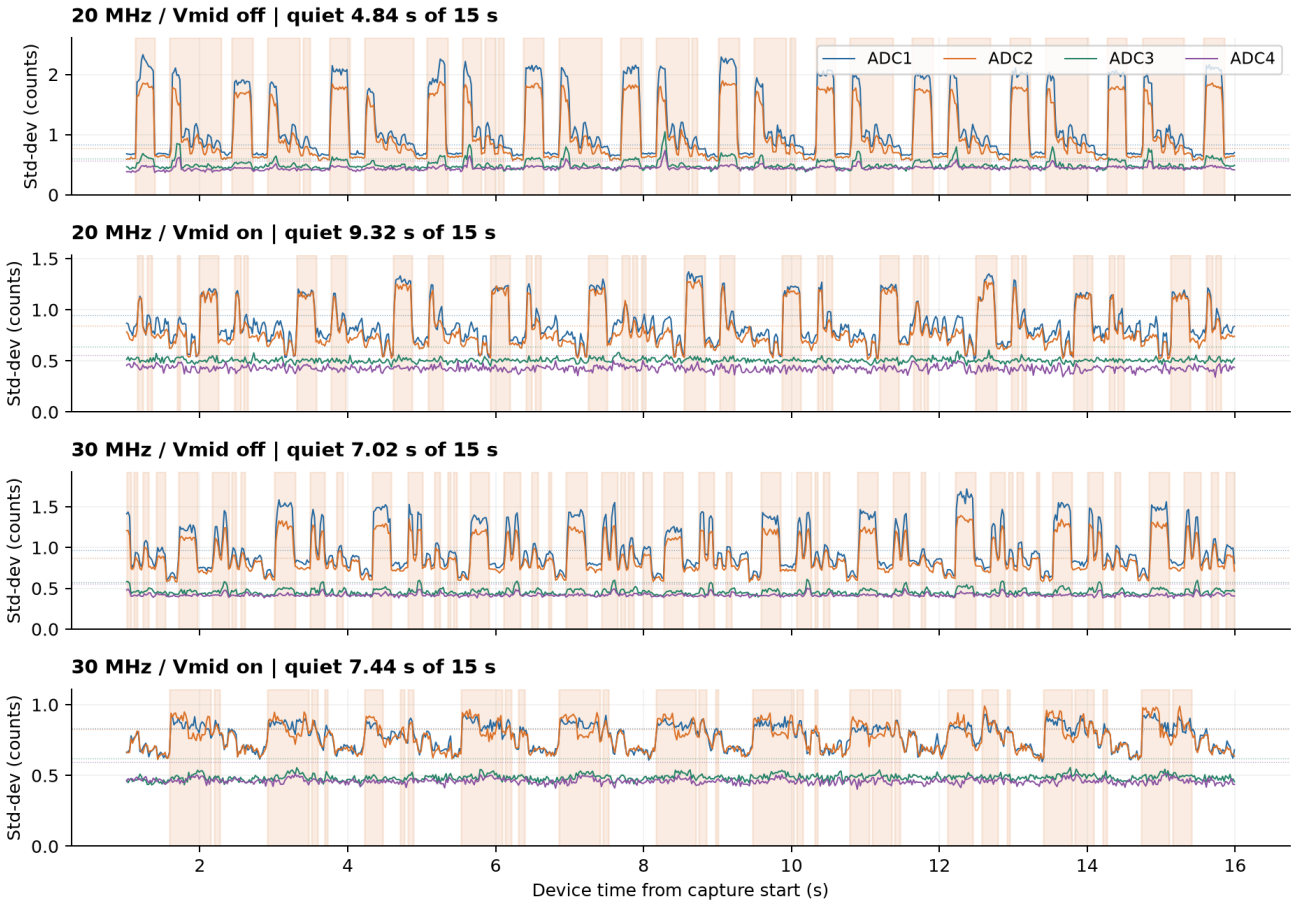
Increasing SPI 20 to 30 MHz gains **14.8%** received rate with Vmid off.

Increasing SPI 20 to 30 MHz gains **15.3%** received rate with Vmid on.

Rates are sustained received sweeps per device elapsed time; one value/channel/sweep. Sweep, period and overhead columns are medians and need not add exactly. Error bars show the three-repetition range. 30 MHz is an experimental setting beyond the ADS7953 rated 20 MHz SPI interface; transport checks alone do not validate analog accuracy.

Quiet-period selection and burst exclusion

Solid curves: median of per-channel sample standard deviations within each ADC, in complete 20 ms bins. Shaded bins are excluded from all ADCs; dotted lines show the thresholds. The first repetition of each setting is shown. All 12 masks and thresholds are saved with the analysis.



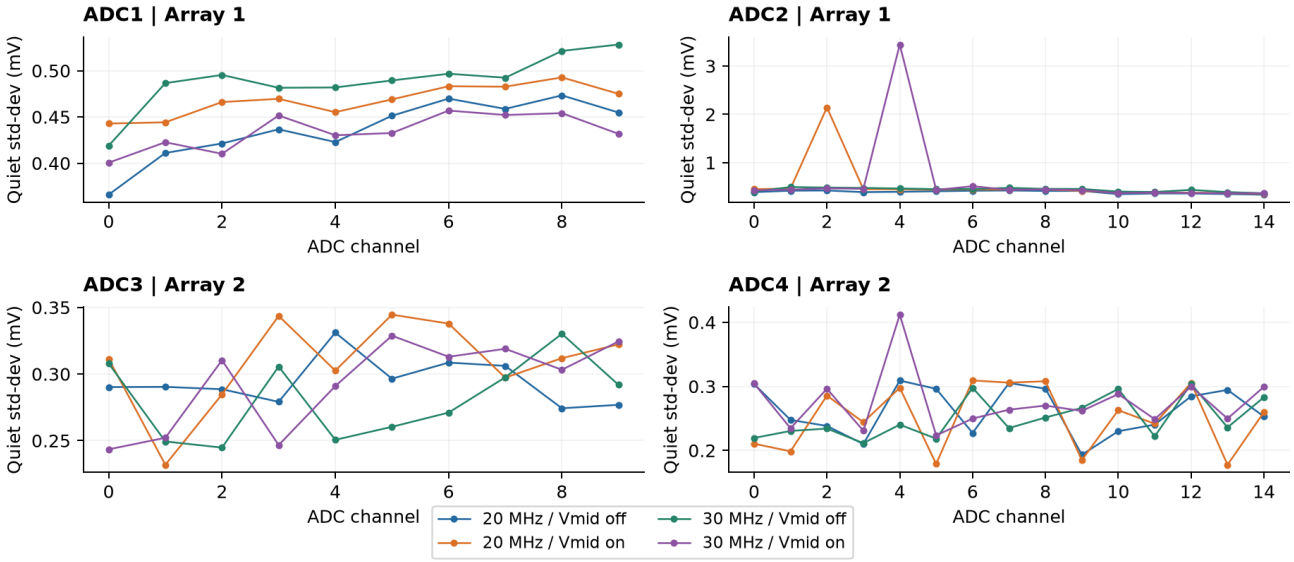
SPI / Vmid	Retained quiet time per capture (s)	Rejected time (%)
20 MHz / Vmid off	4.18 [3.82-4.84]	72.1
20 MHz / Vmid on	9.32 [8.28-10.40]	37.9
30 MHz / Vmid off	7.02 [5.60-7.28]	53.2
30 MHz / Vmid on	7.52 [7.44-8.32]	49.9

Per ADC, threshold = max(P20 of the envelope + 0.15 counts, median of its lower 40% + 4 x 1.4826 x MAD of that lower subset). Reject a bin above any ADC threshold, plus one neighboring bin on each side. This conservative common mask excludes bursts and their transitions without clipping individual values. It defines quiet periods operationally; it is not a synchronized measurement of the charger or rail voltage.

Noise during retained quiet periods

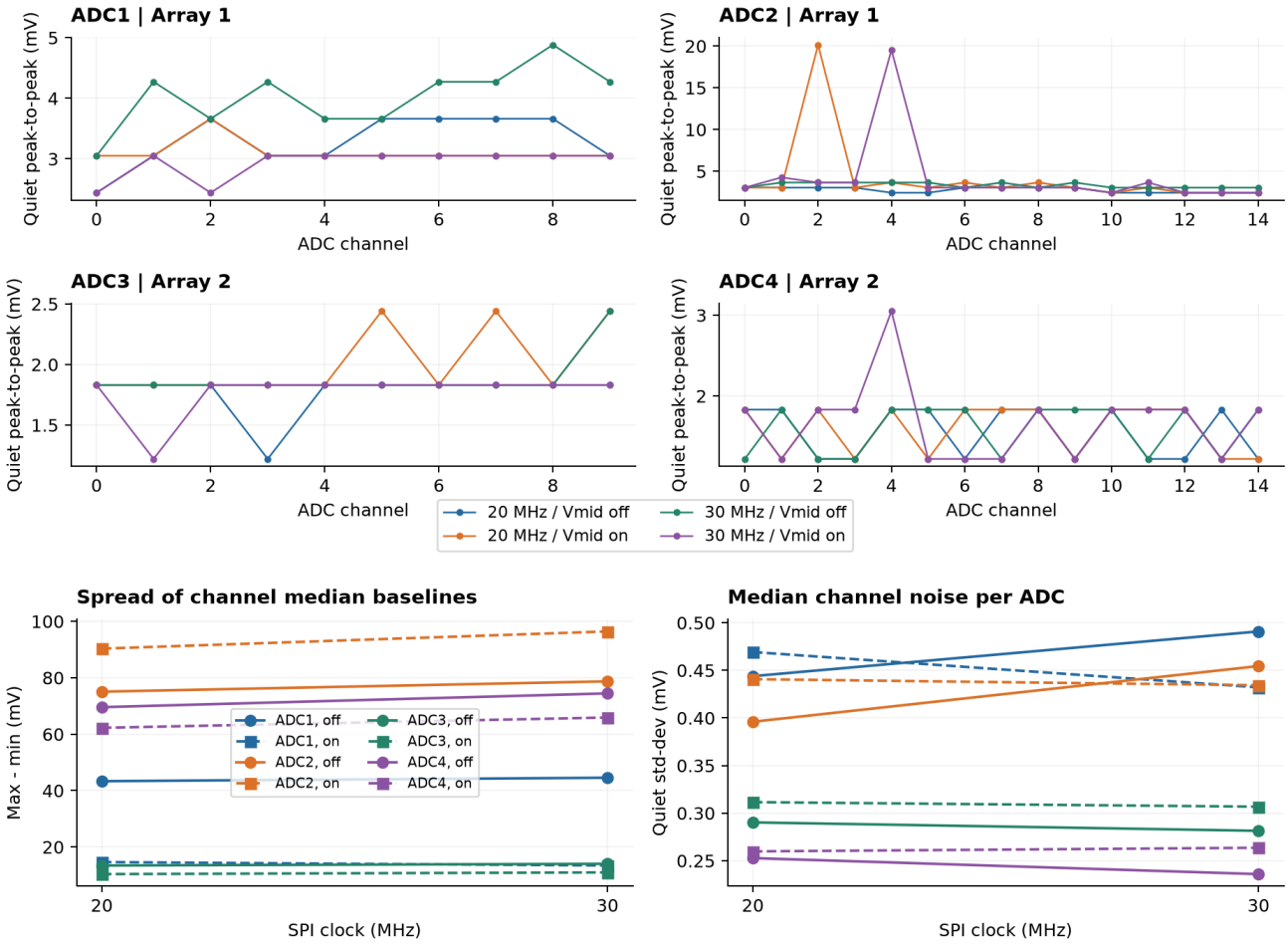
Each cell shows median / maximum across channels, after taking the median of three repetitions per channel. Standard deviation uses ddof=1; peak-to-peak is the actual maximum minus minimum over retained quiet samples. No smoothing or detrending. V = nominal counts x 2.5/4096; one count = 0.000610352 V.

MHz / Vmid	ADC	Std-dev counts median / max	Std-dev V median / max	Peak-peak counts median / max	Peak-peak V median / max
20 / off	ADC1	0.727 / 0.775	0.000444 / 0.000473	5.5 / 6.0	0.003357 / 0.003662
20 / off	ADC2	0.649 / 0.694	0.000396 / 0.000423	5.0 / 5.0	0.003052 / 0.003052
20 / off	ADC3	0.476 / 0.542	0.000290 / 0.000331	3.0 / 3.0	0.001831 / 0.001831
20 / off	ADC4	0.414 / 0.506	0.000253 / 0.000309	3.0 / 3.0	0.001831 / 0.001831
20 / on	ADC1	0.769 / 0.807	0.000469 / 0.000493	5.0 / 6.0	0.003052 / 0.003662
20 / on	ADC2	0.722 / 3.510	0.000441 / 0.002143	5.0 / 33.0	0.003052 / 0.020142
20 / on	ADC3	0.510 / 0.564	0.000312 / 0.000345	3.0 / 4.0	0.001831 / 0.002441
20 / on	ADC4	0.426 / 0.507	0.000260 / 0.000309	3.0 / 3.0	0.001831 / 0.001831
30 / off	ADC1	0.804 / 0.866	0.000491 / 0.000528	7.0 / 8.0	0.004272 / 0.004883
30 / off	ADC2	0.745 / 0.809	0.000455 / 0.000494	5.0 / 6.0	0.003052 / 0.003662
30 / off	ADC3	0.461 / 0.541	0.000281 / 0.000330	3.0 / 4.0	0.001831 / 0.002441
30 / off	ADC4	0.387 / 0.498	0.000236 / 0.000304	3.0 / 3.0	0.001831 / 0.001831
30 / on	ADC1	0.708 / 0.748	0.000432 / 0.000457	5.0 / 5.0	0.003052 / 0.003052
30 / on	ADC2	0.712 / 5.643	0.000434 / 0.003444	5.0 / 32.0	0.003052 / 0.019531
30 / on	ADC3	0.503 / 0.539	0.000307 / 0.000329	3.0 / 3.0	0.001831 / 0.001831
30 / on	ADC4	0.432 / 0.675	0.000264 / 0.000412	3.0 / 5.0	0.001831 / 0.003052



Graphs use mV for readability; full counts/V are in per_channel_metrics.csv. Quiet std-dev includes slow variation. Peak-to-peak depends on sample count and retained record length; a larger extreme at 30 MHz does not alone establish higher noise.

Quiet peak-to-peak and per-ADC tradeoffs

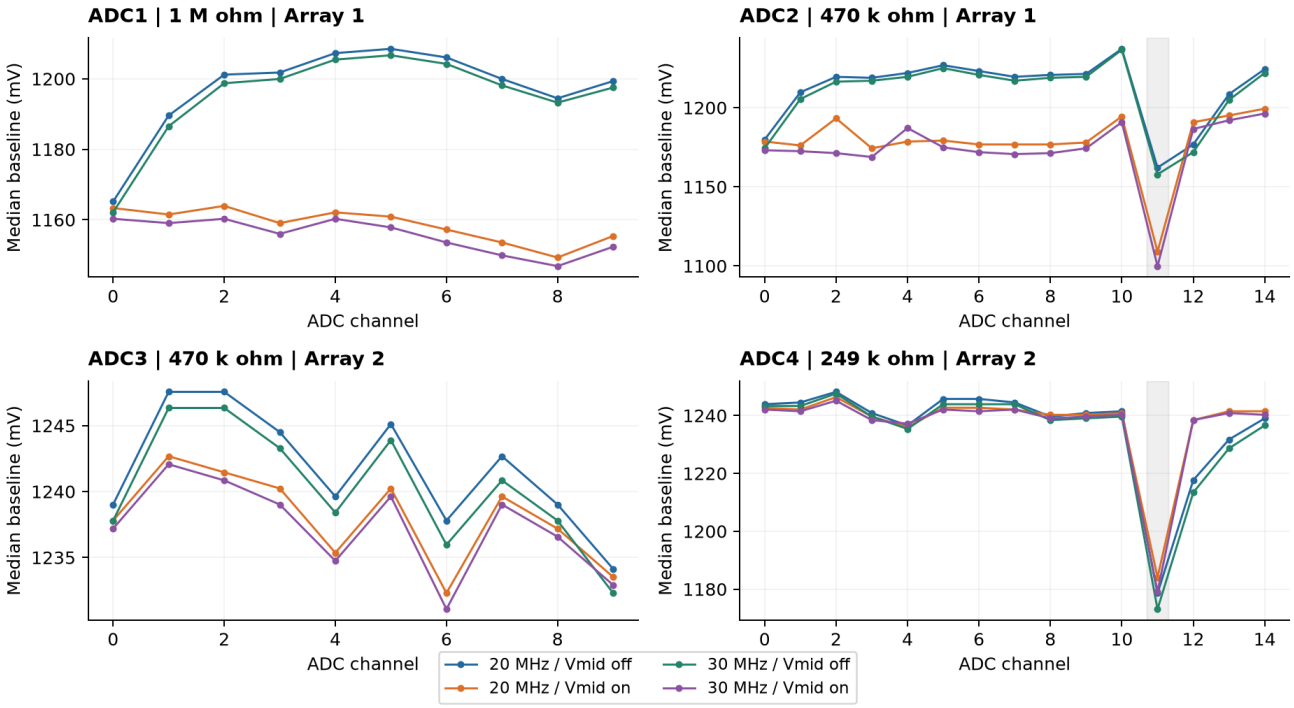


Solid lines/circles: Vmid off. Dashed lines/squares: Vmid on. Spread compares channel median baselines within each ADC; noise compares variations of each channel around its own quiet mean. These are different measurements. Array 1 is disconnected, so its baseline/noise must not be treated as a matched sensor-array control for Array 2.

Persistent individual-channel variations: 20 MHz / Vmid on, ADC2 ch2: std-dev 3.510 counts (0.002143 V), peak-to-peak 33 counts (0.020142 V); 30 MHz / Vmid on, ADC2 ch4: std-dev 5.643 counts (0.003444 V), peak-to-peak 32 counts (0.019531 V). Both recur in all three repetitions. The common charger-burst mask leaves these channel-specific variations visible; they are not removed as individual outliers. Their source is unresolved by this comparison.

Baseline of every sampled channel

Median quiet baseline per channel, then median across three repetitions. Channel 11 on ADC2 and ADC4 is highlighted. ADC1/ADC3 scan channels 0-9; ADC2/ADC4 scan channels 0-14. Channel 15 Vmid conversions are internal parking/insertion measurements and are not transmitted in this production comparison.



ADC / Array	Shared MXO bias	Connector channels	Present sensor strip
ADC1 / 1	1 M ohm	0-9	Disconnected
ADC2 / 1	470 k ohm	0-14	Disconnected
ADC3 / 2	470 k ohm	0-9	Attached: PZT6 / PZT7
ADC4 / 2	249 k ohm	0-14	Attached: PZT1 / PZT3 / PZT5

Bias resistance is at the shared MXO/op-amp node, not separately on every individual ADC/MUX channel. Different ADCs have different resistance and connection conditions. Differences between ADCs are therefore descriptive, not an isolated test of one factor.

Baseline spread, integrity and conclusions

MHz / Vmid	ADC	Max-min counts	Max-min V	Baseline std counts	Baseline std V
20 / off	ADC1	71.0	0.043335	19.72	0.012035
20 / off	ADC2	123.0	0.075073	33.65	0.020541
20 / off	ADC3	22.0	0.013428	6.96	0.004248
20 / off	ADC4	114.0	0.069580	27.72	0.016921
20 / on	ADC1	24.0	0.014648	7.35	0.004486
20 / on	ADC2	148.0	0.090332	33.29	0.020320
20 / on	ADC3	17.0	0.010376	5.35	0.003266
20 / on	ADC4	102.0	0.062256	23.62	0.014419
30 / off	ADC1	73.0	0.044556	20.34	0.012413
30 / off	ADC2	129.0	0.078735	35.51	0.021673
30 / off	ADC3	23.0	0.014038	7.22	0.004405
30 / off	ADC4	122.0	0.074463	29.70	0.018129
30 / on	ADC1	22.0	0.013428	7.45	0.004545
30 / on	ADC2	158.0	0.096436	35.36	0.021582
30 / on	ADC3	18.0	0.010986	5.47	0.003341
30 / on	ADC4	108.0	0.065918	25.28	0.015428

Baseline std is population standard deviation across channel median baselines (ddof=0), not time-domain noise. Max-min is the highest minus lowest channel median within one ADC.

Integrity: all 12 captures passed benchmark decoding, independent wire-sample comparison, timestamp/replay checks and acquired/sent/discarded reconciliation. 2,440,337 received sweeps; 164 whole-sweep USB omissions (0.0067% of acquired sweeps, including warm-up). No increments of ADC, returned-channel, transfer-timeout, DMA/LPSPI-start or USB-write error counters. Actual sampling periods over 1 ms: 0.

Rate: 20 MHz: Vmid insertion lowers received rate by 59.9%. 30 MHz: Vmid insertion lowers received rate by 59.8%. The added conversion history has a larger rate cost than the benefit from increasing SPI to 30 MHz.

Channel 11: ADC2 channel 11 minus mean of channels 10/12: 20/off: -73.5 counts (-44.86 mV); 20/on: -137.0 counts (-83.62 mV); 30/off: -76.0 counts (-46.39 mV); 30/on: -145.5 counts (-88.81 mV).

Channel 11: ADC4 channel 11 minus mean of channels 10/12: 20/off: -83.5 counts (-50.96 mV); 20/on: -91.0 counts (-55.54 mV); 30/off: -87.5 counts (-53.41 mV); 30/on: -98.5 counts (-60.12 mV).

Scope: this is a quiet-baseline comparison under the stated wiring, not a new ghosting test. Earlier manual pressing showed Vmid insertion reduces forward ghosting; the current baseline and noise measurements alone do not establish ghosting. Keep the tested operating choice separate from analog settling and calibrated-voltage acceptance.

Evidence retained: raw binary captures and SHA-256 hashes, command/status logs, 12 repetition metrics and quiet masks, 200 channel-level rows, and 16 ADC summary rows. No firmware or wire-format changes were made. Board left stopped after testing.